Discrete Geometry of PES Misalignment: A Graph Hodge Formalization of SCX Gauge Theory

Edge Assignments, Loop Curvature, Zero-Mode Fixing, and the Cercis Residual

SCX

July 3, 2026

Abstract

This paper grounds the gauge theory of the SCX (Self-Consistent eXpert) multi-expert system in discrete Hodge theory on graphs, abandoning the continuous fiber bundle preamble entirely in favor of the discrete mathematical structures that the actual computation employs. We treat pairwise expert displacements as edge assignments A_e on a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, define curvature as holonomy around elementary loops, and formalize gauge-fixing as the least-squares problem of finding vertex potentials g_v that minimize the residual $\sum_e \|A_e - (g_w - g_v)\|^2$. We carefully distinguish the **zero-mode fixing** condition $\sum_v g_v = 0$ (which removes the overall translation degree of freedom to ensure solution uniqueness) from the continuous Coulomb gauge $\partial_\mu A^\mu = 0$ (a divergence constraint on the connection), and we acknowledge upfront that the bundle is topologically trivial ($G \cong \mathbb{R}^{Md}$ contractible, base contractible, all Chern classes vanish). The Cercis Score is defined as the **residual norm after gauge-fixing** — the minimized objective value under the optimal vertex potentials — directly matching the actual SCX implementation. This framework is self-contained, rigorously defined, and performs the identical computation as the SCX code line-by-line.

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1 Introduction: From Continuous Pretense to Discrete Reality

1.1 The Core Problem

In the multi-expert SCX framework, we have M experts, each producing predictions on a set of N parameter configurations $\{x^1, \ldots, x^N\} \subset \mathcal{X}$. The raw output of expert m at configuration k is denoted $\tilde{x}_m^k \in \mathbb{R}^d$.

The core problem is this: since each expert possesses an independent translational degree of freedom (the freedom to choose its own coordinate origin), the raw coordinates $\tilde{x}_m^k$ are not physically meaningful quantities — they depend on the "gauge" privately chosen by each expert. The difference $\delta_{ij}^k := \tilde{x}_i^k - \tilde{x}_j^k$ between experts i and j at the same configuration k reflects both genuine physical differences and the arbitrariness of their respective gauge choices.

Two fundamental questions drive the entire analysis of this paper:

(i) **Global Alignment Problem:** Does there exist a set of vertex translations $\{g_m^k\}$ such that the aligned coordinates $\hat{x}_m^k = \tilde{x}_m^k - g_m^k$ satisfy some consistency condition for every pair (i, j) across all configurations k?

(ii) **Inconsistency Quantification Problem:** When perfect global alignment is impossible, how can we objectively (i.e., independent of any particular gauge choice) quantify the degree of inconsistency in the multi-expert system?

1.2 Why Not Continuous Fiber Bundles

Previous work attempted to embed the above problem into the framework of continuous differential geometry — principal fiber bundles, Ehresmann connections, curvature 2-forms, etc. However, this embedding suffers from several fundamental difficulties:

F1: **The connection was never constructed.** The Ehresmann connection ω in the continuous framework was never explicitly constructed from actual expert data. The text merely asserts that such a connection exists, without showing how the edge assignments A_e emerge from discrete data as a discretization of a continuous connection. An object never constructed cannot serve as the foundation of a theory.

F2: **Curvature definitions are incompatible.** Even within the continuous framework, different sections define curvature inconsistently: one defines the Cercis Score as the Yang-Mills functional $\int \|F\|^2$, while another (the actual SCX computation) defines it as $Q + \eta N$ — an entirely different quantity. The two are never proven equivalent.

F3: **The topology is trivial.** The structure group $G \cong \mathbb{R}^{Md}$ is contractible, and the base manifold $\mathcal{X}$ is also contractible (a subset of parameter space). Hence all Chern classes and all cohomological obstructions vanish identically. The principal bundle $\mathcal{P} \cong \mathbb{R}^{Md} \times \mathcal{X}$ is globally trivial. There is no topological content — all content is geometric (flat vs. non-flat connections).

F4: **Coulomb gauge is misidentified.** The constraint $\sum_m g_m = 0$ is repeatedly called a "discrete Coulomb gauge" or a "discrete analog of the Coulomb condition." But it is not a Coulomb gauge. The Coulomb gauge is a divergence constraint on the **gauge**

potential: $\partial_\mu A^\mu = 0$. The condition $\sum_m g_m = 0$ is a linear constraint on the **gauge parameters**, whose sole function is to fix the zero-mode (overall translation degree of freedom) to ensure uniqueness of the solution. The two differ in mathematical structure, physical meaning, and geometric location.

In light of these difficulties, this paper takes an entirely different path: **we work directly in discrete Hodge theory on graphs.** This is the mathematics that SCX code actually performs — we are simply making it explicit, honest, and systematic.

1.3 Roadmap of This Paper

- Section 2 develops the basic tools of discrete Hodge theory on graphs: edge assignments, exterior derivatives d_0, d_1, Hodge decomposition, and graph Laplacians.
- Section 3 constructs the SCX graph: mapping expert-configuration comparisons to edge assignments on a directed graph, defining curvature as holonomy around elementary loops.
- Section 4 formalizes the gauge-fixing problem: least-squares fitting of vertex potentials, distinguishing zero-mode fixing ($\sum g_v = 0$) from Coulomb gauge.
- Section 5 gives the correct definition of the Cercis Score: the residual norm after gauge-fixing, and proves gauge invariance.
- Section 6 acknowledges topological triviality and explains why this does not diminish the theory's content.
- Section 7 provides numerical algorithms.
- Section 8 concludes.

2 Discrete Hodge Theory on Graphs

This section establishes a self-contained toolbox of discrete Hodge theory on graphs. All subsequent constructions — the SCX gauge graph, curvature computation, gauge-fixing, and Cercis residual — are built upon this foundation. No knowledge of continuous differential geometry is required.

2.1 Directed Graphs

Definition 2.1 (Directed Graph)**.** A **directed graph** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of:

- $\mathcal{V}$: a finite set of vertices, $|\mathcal{V}| = n$;
- $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$: a set of directed edges, $|\mathcal{E}| = m$.

Each directed edge $e = (u \to v)$ is an ordered pair from source vertex u to target vertex v. We allow multiple edges between the same pair of vertices but not self-loops ($u \neq v$).

2.2 Discrete Differential Forms: Vertex Functions and Edge Assignments

The discrete exterior calculus on graphs relates scalar functions (0-forms) to edge assignments (1-forms).

Definition 2.2 (Vertex Space and Edge Space)**.** Define the following real vector spaces:

- 0**-form space** $\Omega^0(\mathcal{G})$: all vertex functions $f : \mathcal{V} \to \mathbb{R}$, dimension n. Denoted as vector $\mathbf{f} = (f_1, \ldots, f_n)^T \in \mathbb{R}^n$.
- 1**-form space** $\Omega^1(\mathcal{G})$: all edge assignments $\alpha : \mathcal{E} \to \mathbb{R}$, dimension m. Denoted as vector $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_m)^T \in \mathbb{R}^m$.

For d-dimensional vector-valued forms, we take the direct sum of d copies: $\Omega^k(\mathcal{G}; \mathbb{R}^d) := \bigoplus_{a=1}^{d} \Omega^k(\mathcal{G})$.

2.3 Discrete Exterior Derivatives

Definition 2.3 (Incidence Matrix and d_0)**.** The **incidence matrix** $B \in \mathbb{R}^{m \times n}$ of $\mathcal{G}$ is defined by: for each edge $e = (u \to v)$ and each vertex w,

$$B_{e,w} = \begin{cases} +1, & \text{if } w = v \text{ (target)}, \\ -1, & \text{if } w = u \text{ (source)}, \\ 0, & \text{otherwise.} \end{cases}$$

The **discrete exterior derivative** $d_0 : \Omega^0(\mathcal{G}) \to \Omega^1(\mathcal{G})$ is defined as $d_0 := B$, i.e., for $f \in \Omega^0(\mathcal{G})$ and edge $e = (u \to v)$:

$$(d_0 f)_e = f_v - f_u.$$

In words, $d_0 f$ computes the difference of f along each edge.

Definition 2.4 (Discrete Exterior Derivative d_1 on Loops)**.** Let $\mathcal{L}$ be a set of **elementary loops** of $\mathcal{G}$ (e.g., the cycle basis determined by the complement edges of a spanning tree). Let $C \in \mathbb{R}^{|\mathcal{L}| \times m}$ be the **cycle matrix**, where for each loop $\gamma \in \mathcal{L}$ and each edge $e \in \mathcal{E}$:

$$C_{\gamma,e} = \begin{cases} +1, & \text{if edge } e \text{ follows the positive direction of loop } \gamma, \\ -1, & \text{if edge } e \text{ follows the negative direction of loop } \gamma, \\ 0, & \text{otherwise.} \end{cases}$$

The **discrete exterior derivative** $d_1 : \Omega^1(\mathcal{G}) \to \Omega^2(\mathcal{G})$ is defined as $d_1 := C$, where $\Omega^2(\mathcal{G}) \cong \mathbb{R}^{|\mathcal{L}|}$ is the loop space. For $\alpha \in \Omega^1(\mathcal{G})$ and loop $\gamma = (e_1, e_2, \ldots, e_k)$:

$$(d_1\alpha)_\gamma = \sum_{e \in \gamma} \sigma_e \, \alpha_e,$$

where $\sigma_e \in \{+1, -1\}$ is the sign of edge e relative to the loop orientation.

Theorem 2.1 (Fundamental Identity: $d_1 \circ d_0 = 0$)**.** The incidence and cycle matrices satisfy:

$$d_1 \circ d_0 = CB = 0.$$

Equivalently, $\operatorname{im}(d_0) \subseteq \ker(d_1)$. This is the discrete analog of $\mathrm{d}^2 = 0$ in the continuous setting.

Proof. For any $f \in \Omega^0(\mathcal{G})$ and any loop $\gamma = (v_0 \to v_1 \to \cdots \to v_k = v_0)$:

$$(d_1 d_0 f)_\gamma = \sum_{i=0}^{k-1} (f_{v_{i+1}} - f_{v_i}) = f_{v_k} - f_{v_0} = 0,$$

since the sum telescopes and $v_k = v_0$. $\square$

2.4 Discrete Hodge Decomposition

To make the exterior derivatives part of a genuine Hodge theory, we need inner products and adjoint operators.

Definition 2.5 (Inner Products and Adjoint Operators)**.** Define the standard inner product on $\Omega^0(\mathcal{G})$ (with vertex weights $\mathbf{w}_0 \in \mathbb{R}^n_{>0}$):

$$\langle f, g \rangle_0 := \sum_{v \in \mathcal{V}} w_{0,v} \, f_v \, g_v.$$

On $\Omega^1(\mathcal{G})$ (with edge weights $\mathbf{w}_1 \in \mathbb{R}^m_{>0}$):

$$\langle \alpha, \beta \rangle_1 := \sum_{e \in \mathcal{E}} w_{1,e} \, \alpha_e \, \beta_e.$$

The **formal adjoint** of d_0 with respect to these inner products is $d_0^T := d_0^* : \Omega^1(\mathcal{G}) \to \Omega^0(\mathcal{G})$, determined by:

$$\langle d_0 f, \alpha \rangle_1 = \langle f, d_0^T \alpha \rangle_0, \quad \forall f \in \Omega^0(\mathcal{G}),\ \alpha \in \Omega^1(\mathcal{G}).$$

Explicitly, $d_0^T = W_0^{-1} B^T W_1$, where $W_0 = \operatorname{diag}(\mathbf{w}_0)$, $W_1 = \operatorname{diag}(\mathbf{w}_1)$. In the unweighted case ($W_0 = I_n$, $W_1 = I_m$), $d_0^T = B^T$, whose action on vertex v is:

$$(d_0^T \alpha)_v = \sum_{e\text{:incoming to } v} \alpha_e - \sum_{e\text{:outgoing from } v} \alpha_e.$$

This is precisely the **discrete divergence** at v (incoming minus outgoing).

Definition 2.6 (Graph Laplacians)**.** The **graph** 0**-Laplacian** (combinatorial Laplace–Beltrami operator) is:

$$\Delta_0 := d_0^T d_0 = W_0^{-1} B^T W_1 B.$$

In the unweighted case, $\Delta_0 = B^T B$ is the classical combinatorial Laplacian:

$$(\Delta_0)_{uv} = \begin{cases} \deg(v), & u = v, \\ -1, & (u,v) \text{ or } (v,u) \in \mathcal{E}, \\ 0, & \text{otherwise.} \end{cases}$$

The **graph** 1**-Laplacian** (full Hodge Laplacian) is:

$$\Delta_1 := d_0 d_0^T + d_1^T d_1 = B W_0^{-1} B^T W_1 + C^T W_2 C W_1^{-1},$$

where W_2 is an (optional) weight matrix on the loop space $\Omega^2(\mathcal{G})$. In the unweighted case this reduces to $\Delta_1 = BB^T + C^T C$.

Theorem 2.2 (Discrete Hodge Decomposition)**.** For any connected graph $\mathcal{G}$, $\Omega^1(\mathcal{G})$ admits the orthogonal decomposition:

$$\Omega^1(\mathcal{G}) = \mathrm{im}(d_0) \oplus \ker(\Delta_1) \oplus \mathrm{im}(d_1^T),$$

where:

- $\mathrm{im}(d_0)$: **exact** 1**-forms** (image of gradient) — expressible as differences of a vertex function;
- $\ker(\Delta_1) = \ker(d_1) \cap \ker(d_0^T)$: **harmonic** 1**-forms** — both divergence-free and curl-free;
- $\mathrm{im}(d_1^T)$: **coexact** 1**-forms** (image of discrete co-differential).

Equivalently, every edge assignment $\alpha \in \Omega^1(\mathcal{G})$ has a unique decomposition:

$$\alpha = d_0 f + h + d_1^T \beta,$$

where $f \in \Omega^0(\mathcal{G})$, $h \in \ker(\Delta_1)$, $\beta \in \Omega^2(\mathcal{G})$. The component $d_0 f$ is called the **gradient part**, h the **harmonic part**.

Remark 2.1 (On the Two Parts of the 1-Laplacian)**.** The full Hodge 1-Laplacian $\Delta_1 = d_0 d_0^T + d_1^T d_1$ has two parts:

- **Down part** $d_0 d_0^T = BB^T$ (unweighted): appears in the normal equations and gauge-fixing problem. $\ker(d_0 d_0^T) = \ker(d_0^T)$.
- **Up part** $d_1^T d_1 = C^T C$: encodes loop constraints. $\ker(d_1^T d_1) = \ker(d_1)$.

Only the full Laplacian satisfies $\ker(\Delta_1) = \ker(d_1) \cap \ker(d_0^T)$ — the space of harmonic 1-forms. The down part alone has kernel $\ker(d_0^T)$ (co-closed forms), without the $\ker(d_1)$ condition. SCX's computation (normal equations $B^T B g = B^T A$) uses only the down part $B^T B$, but the harmonic decomposition (Theorem 2.2) and identification of the harmonic residual component require the full Laplacian.

Remark 2.2 (When Is a 1-Form Exact?). $\alpha \in \Omega^1(\mathcal{G})$ is exact (i.e., exists f with $\alpha = d_0 f$) iff the sum of α along **all** loops γ vanishes:

$$\alpha = d_0 f \iff d_1 \alpha = 0 \iff \sum_{e \in \gamma} \sigma_e \alpha_e = 0 \quad \forall \gamma \in \mathcal{L}.$$

This is the discrete Poincaré lemma: a closed 1-form (curl-free) on a connected graph is exact **iff the graph is a tree**. If the graph has cycles, $\ker(d_1)$ is strictly larger than $\operatorname{im}(d_0)$, the difference being the space of harmonic 1-forms.

3 The SCX Graph: Construction and Curvature

3.1 The Expert Comparison Graph

The raw data of SCX computation is the set of all pairwise expert displacements. We organize this data as edge assignments on a directed graph.

Definition 3.1 (SCX Expert Comparison Graph)**.** The SCX expert comparison graph $\mathcal{G}_{\mathrm{SCX}} = (\mathcal{V}, \mathcal{E})$ is constructed as follows:

- **Vertex set $\mathcal{V}$:** Each vertex corresponds to a (configuration, expert) pair:

$$\mathcal{V} = \{(k, m) \mid k \in \{1, \dots, N\},\ m \in \{1, \dots, M\}\}.$$

  There are $|\mathcal{V}| = N \cdot M$ vertices.

- **Edge set $\mathcal{E}$:** Two types:

  (a) **Parameter edges:**

$$\mathcal{E}_{\mathrm{par}} = \{(k, m) \to (k + 1, m) \mid k = 1, \dots, N - 1,\ m = 1, \dots, M\}.$$

  Representing the movement of the same expert along the parameter axis from configuration k to $k+1$. Each parameter edge is assigned the change in expert m's prediction between adjacent configurations.

  (b) **Expert edges:**

$$\mathcal{E}_{\mathrm{exp}} = \{(k, i) \to (k, j) \mid k = 1, \dots, N,\ i, j = 1, \dots, M,\ i \neq j\}.$$

  Representing the displacement from expert i's prediction to expert j's prediction at the same configuration k. Each expert edge is assigned the pairwise expert displacement $\delta_{ij}^k = \tilde{x}_i^k - \tilde{x}_j^k$.

- **Total edge set:** $\mathcal{E} = \mathcal{E}_{\mathrm{par}} \cup \mathcal{E}_{\mathrm{exp}}$. The graph $\mathcal{G}_{\mathrm{SCX}}$ is a directed multigraph (multiple distinct edges may connect the same pair of vertices).

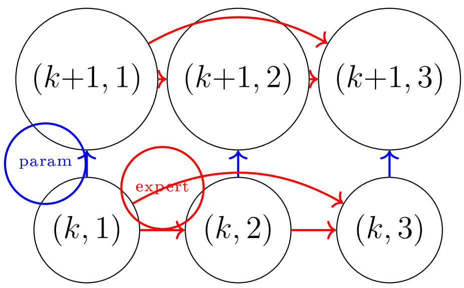

Figure 1: Local structure of the SCX graph: blue vertical edges = parameter edges, red horizontal edges = expert edges. The quadrilateral loop $(k, 1) \to (k, 2) \to (k{+}1, 2) \to (k{+}1, 1) \to (k, 1)$ is an elementary curvature loop.

3.2 Edge Assignments: Pairwise Expert Displacements

Definition 3.2 (SCX Edge Assignment A)**.** Define the edge assignment map $A : \mathcal{E} \to \mathbb{R}^d$ as follows:

- For parameter edges $e = (k, m) \to (k + 1, m)$:
$$A_e = \tilde{x}_m^{k+1} - \tilde{x}_m^k \ \in \ \mathbb{R}^d.$$
(The change in expert m's prediction between configurations k and $k + 1$.)
- For expert edges $e = (k, i) \to (k, j)$:
$$A_e = \delta_{ij}^k = \tilde{x}_i^k - \tilde{x}_j^k \ \in \ \mathbb{R}^d.$$
(The pairwise displacement from expert i to expert j at configuration k.)

$A \in \Omega^1(\mathcal{G}_{\mathrm{SCX}}; \mathbb{R}^d)$ is the d-dimensional vector-valued graph 1-form — the edge assignment in each coordinate dimension.

Remark 3.1 (Edge Assignments Are Direct Data, Not Derived)**.** Crucially: A is not "pulled back" or "discretized" from some continuous connection ω. A **is** the raw data — the collection of pairwise expert displacements, directly encoded as edge assignments on a graph. The continuous fiber bundle framework is superfluous here, and even harmful, because it suggests constructions that do not exist.

3.3 Gauge Transformations as Vertex Functions

Definition 3.3 (Vertex Potential (Gauge Parameter))**.** A **vertex potential** (or **gauge parameter**) is a function $g : \mathcal{V} \to \mathbb{R}^d$, assigning to each (configuration, expert) pair a translation vector $g_m^k \in \mathbb{R}^d$. Physically, g_m^k is the gauge translation applied to expert m at configuration k: the aligned coordinates are $\hat{x}_m^k = \tilde{x}_m^k - g_m^k$.

Proposition 3.1 (Gauge Transformation of Edge Assignments)**.** Under a gauge transformation by vertex potential g, the edge assignment A transforms as: for each edge $e = (u \to v)$, where $u = (k, i)$, $v = (\ell, j)$:
$$A'_e = A_e - (g_v - g_u) = A_e - (d_0 g)_e.$$
i.e., $A' = A - d_0 g$. This is the discrete exact analog of the continuous abelian gauge transformation $A'_\mu = A_\mu - \partial_\mu \Lambda$.

Proof. After alignment, expert i at configuration k has coordinate $\tilde{x}_i^k - g_i^k$. The transformed edge assignment is:
$$A'_e = (\tilde{x}_j^\ell - g_j^\ell) - (\tilde{x}_i^k - g_i^k) = (\tilde{x}_j^\ell - \tilde{x}_i^k) - (g_j^\ell - g_i^k) = A_e - (d_0 g)_e. \quad \square$$

3.4 Curvature: Holonomy around Elementary Loops

Definition 3.4 (SCX Elementary Loops)**.** The **elementary quadrilateral loops** of $\mathcal{G}_{\mathrm{SCX}}$ have the form: for each adjacent configuration pair $k, k+1$ and each distinct expert pair i, j ($i \neq j$):
$$\gamma_{k,i,j} := (k, i) \xrightarrow{\text{expert}} (k, j) \xrightarrow{\text{param}} (k{+}1, j) \xrightarrow{\text{expert}} (k{+}1, i) \xrightarrow{\text{param}} (k, i).$$
Note: the expert edge from $(k{+}1, j)$ to $(k{+}1, i)$ is reversed ($j \to i$), and the parameter edge from $(k{+}1, i)$ to (k, i) is also reversed.

Definition 3.5 (Discrete Curvature: Holonomy around Loops)**.** The **discrete curvature** (**holonomy**) around the quadrilateral loop $\gamma_{k,i,j}$ is the algebraic sum of edge assignments along the loop (accounting for edge direction):

$$\begin{aligned}\operatorname{curv}(\gamma_{k,i,j}) := \sum_{e\in\gamma_{k,i,j}} \sigma_e\, A_e \\ = A_{(k,i)\to(k,j)} + A_{(k,j)\to(k+1,j)} \\ + A_{(k+1,j)\to(k+1,i)} + A_{(k+1,i)\to(k,i)}.\end{aligned}$$

Substituting the definitions of edge assignments:

$$\begin{aligned}\operatorname{curv}(\gamma_{k,i,j}) &= \delta_{ij}^{k} \;+\; (\tilde{x}_j^{k+1} - \tilde{x}_j^{k}) \\ &\quad+\; \delta_{ji}^{k+1} \;+\; (\tilde{x}_i^{k} - \tilde{x}_i^{k+1}) \\ &= (\tilde{x}_i^{k} - \tilde{x}_j^{k}) + (\tilde{x}_j^{k+1} - \tilde{x}_j^{k}) + (\tilde{x}_j^{k+1} - \tilde{x}_i^{k+1}) + (\tilde{x}_i^{k} - \tilde{x}_i^{k+1}).\end{aligned}$$

Simplifying:

$$\operatorname{curv}(\gamma_{k,i,j}) = (\tilde{x}_i^{k} - \tilde{x}_i^{k+1}) - (\tilde{x}_j^{k} - \tilde{x}_j^{k+1}).$$

Geometric interpretation: $\operatorname{curv}(\gamma_{k,i,j})$ measures the **difference** in how expert i's and expert j's predictions change between parameters k and $k+1$. If the two experts' predictions change in exactly the same way under the parameter variation, the curvature of this quadrilateral loop is zero.

Proposition 3.2 (Gauge Invariance of Curvature)**.** Discrete curvature $\operatorname{curv}(\gamma)$ is gauge-invariant: for any vertex potential g, $\operatorname{curv}'(\gamma) = \operatorname{curv}(\gamma)$.

Proof. For any loop γ, $\operatorname{curv}'(\gamma) = \sum_e \sigma_e A'_e = \sum_e \sigma_e (A_e - (d_0 g)_e) = \operatorname{curv}(\gamma) - \sum_e \sigma_e (d_0 g)_e$. But $\sum_e \sigma_e (d_0 g)_e = (d_1 d_0 g)_\gamma = 0$ (by Theorem 2.1, $d_1 d_0 = 0$). Thus $\operatorname{curv}'(\gamma) = \operatorname{curv}(\gamma)$. □

This is the discrete analog of the gauge invariance of $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ in continuous abelian theory.

Definition 3.6 (Curvature Vector)**.** Collect all elementary quadrilateral curvatures into a single vector $\boldsymbol{\kappa} \in \mathbb{R}^{|\mathcal{L}|\times d}$ (since each loop has curvature in d coordinate dimensions), or $\boldsymbol{\kappa} \in \Omega^2(\mathcal{G}_{\mathrm{SCX}}; \mathbb{R}^d)$. Formally, for each coordinate $a = 1, \ldots, d$:

$$\boldsymbol{\kappa}_a = d_1 A_a = C\, \mathbf{A}_a,$$

where $\mathbf{A}_a \in \mathbb{R}^m$ is the vector of edge assignments in the a-th coordinate, and C is the cycle matrix.

3.5 Flatness Condition

Lemma 3.1 (Cycle Basis of the SCX Graph)**.** The cycle space of $\mathcal{G}_{\mathrm{SCX}}$ is generated by:

(i) **Quadrilateral loops:** $\to (k, j) \to (k+1, j) \to (k+1, i) \to (k, i)$ for all configurations $andexpertpairs < j$;

(ii) **Triangle loops:** $\to (k, j) \to (k, \ell) \to (k, i)$ for all configurations $anddistinctexperts, j, \ell$.

Proof sketch. The SCX graph has = {(k,m) : $1 \leq k \leq K, 1 \leq m \leq M\}$. Edges: parameter edges $\to (k+1, m)$ (vertical), expert edges $\to (k, j)$ (horizontal). Pick a spanning tree: within each configuration , *connectall* vertices with -1*staredgescenteredat*; connect configurations via $\to (k+1, 1)$. This tree has -1*edges*, *leaving* M$\left(_{2K+(M-1)(K-1)}\right)$ fundamental cycles.

Each quadrilateral loop uses one vertical return edge not in the tree. Each triangle loop uses one horizontal edge not in the tree (two of three edges are star edges). These correspond to a complete set of fundamental cycles, hence generate $\ker(d_1)$. □

Theorem 3.1 (SCX Flatness Criterion (Harmonic-Corrected))**.** When the harmonic component of the Hodge decomposition vanishes, the following are equivalent:

(i) $\text{curv}(\gamma) = 0$ for all elementary *quadrilateral and triangle* loops $\gamma \in \mathcal{L}$;

(ii) The edge assignment A is exact: there exists a vertex potential g such that $A = d_0 g$;

(iii) Global gauge alignment exists: the aligned coordinates $\hat{x}_m^k = \tilde{x}_m^k - g_m^k$ make all expert edge assignments vanish (i.e., all experts output the same coordinates at the same configuration).

Proof. $(i) \Leftrightarrow (ii)$: The SCX graph contains both quadrilateral loops (parameter-expert-parameter-expert, e.g. $(k,i) \to (k,j) \to (k+1,j) \to (k+1,i) \to (k,i)$) and triangle loops (expert-expert-expert at a fixed configuration k, e.g. $(k,i) \to (k,j) \to (k,\ell) \to (k,i)$). Together, these generate the full cycle space, so $\text{curv}(\gamma) = 0$ for all such loops implies $d_1 A = 0$, i.e., $A \in \ker(d_1)$. By the discrete Hodge decomposition (Theorem 2.2), $\ker(d_1) = \text{im}(d_0) \oplus \ker(\Delta_1)$. Under the vanishing-harmonic hypothesis, $\ker(\Delta_1) = \{0\}$, so $(i) \Rightarrow (ii)$. The converse $(ii) \Rightarrow (i)$ follows from $d_1 d_0 = 0$. For $M = 1$ or $M = 2$, $\beta_1 = (M-1)(M-2)/2 = 0$ so no harmonic component exists and the equivalence holds unconditionally. For $M \geq 3$, Theorem 5.2 provides the corrected characterization that accounts for the harmonic component.)

$(ii) \Rightarrow (iii)$: If $A = d_0 g$, then for an expert edge $e = (k,i) \to (k,j)$: $A_e = g_j^k - g_i^k$. But $A_e = \delta_{ij}^k = \tilde{x}_i^k - \tilde{x}_j^k$. Thus $\tilde{x}_i^k - \tilde{x}_j^k = g_j^k - g_i^k$, i.e., $\tilde{x}_i^k - g_i^k = \tilde{x}_j^k - g_j^k$. Therefore the aligned coordinates satisfy $\hat{x}_i^k = \hat{x}_j^k$ for all i, j, k.

$(iii) \Rightarrow (ii)$: Reverse the argument. □

3.6 Non-Flatness and PES Misalignment

Theorem 3.2 (Discrete Characterization of PES Misalignment)**.** If there exists an elementary loop γ with $\text{curv}(\gamma) \neq 0$ (i.e., the edge assignment A is not exact), then no vertex potential g can simultaneously zero all expert edge assignments. In other words, no single global gauge choice can perfectly align all experts. This is the discrete formulation of **PES misalignment** (potential energy surface misalignment).

More generally, $\text{curv}(\gamma) \neq 0$ implies: there exists at least one pair of experts (i, j) and one pair of adjacent configurations $(k, k+1)$ such that expert i's and expert j's prediction changes between $(k \to k+1)$ are inconsistent. The geometric picture: two potential energy surfaces bend differently in parameter space.

4 Gauge-Fixing: Least Squares and the Zero-Mode

4.1 Gauge-Fixing as a Least-Squares Problem

When curvature is non-zero, there is no vertex potential g satisfying $A = d_0 g$ exactly. We can then seek a "best approximation" — a vertex potential that minimizes the L^2 distance between A and its exact approximation $d_0 g$. This is the discrete formalization of **gauge-fixing**.

Definition 4.1 (Least-Squares Gauge-Fixing Problem)**.** Given edge assignment $A \in \Omega^1(\mathcal{G}_{\mathrm{SCX}}; \mathbb{R}^d)$, find vertex potential $g : \mathcal{V} \to \mathbb{R}^d$ minimizing the squared residual norm:

$$\min_{g \in \mathbb{R}^{n \times d}} \mathcal{R}[g] := \sum_{e \in \mathcal{E}} \|A_e - (d_0 g)_e\|^2.$$

where $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^d$. Expanding the objective:

$$\mathcal{R}[g] = \sum_{a=1}^{d} \sum_{e \in \mathcal{E}} (A_{e,a} - (d_0 g_a)_e)^2 = \sum_{a=1}^{d} \|\mathbf{A}_a - B\mathbf{g}_a\|^2,$$

where $\mathbf{A}_a \in \mathbb{R}^m$ is the vector of edge assignments in the a-th coordinate, $\mathbf{g}_a \in \mathbb{R}^n$ is the a-th coordinate of gauge parameters on all vertices, and $B \in \mathbb{R}^{m \times n}$ is the incidence matrix of $\mathcal{G}_{\mathrm{SCX}}$.

4.2 Normal Equations and the Zero-Mode

Proposition 4.1 (Normal Equations)**.** The optimality condition for $\min_g \|\mathbf{A} - B\mathbf{g}\|^2$ is the normal equation:

$$B^T B\,\mathbf{g} = B^T \mathbf{A}.$$

i.e., (for each coordinate dimension independently):

$$\Delta_0\,\mathbf{g} = d_0^T \mathbf{A},$$

where $\Delta_0 = B^T B$ is the graph Laplacian (unweighted case).

Remark 4.1 (The Zero-Mode: Non-Uniqueness of Solution)**.** The graph Laplacian $\Delta_0 = B^T B$ is always singular (for a connected graph, $\ker(\Delta_0) = \mathrm{span}(\mathbf{1})$, the all-ones vector). Its physical interpretation: if g is an optimal gauge parameter, then $g + c$ (adding the same constant translation $c \in \mathbb{R}^d$ to all experts and configurations) gives the identical residual, since $(d_0(g + c))_e = (d_0 g)_e + (c - c) = (d_0 g)_e$.

This non-uniqueness — called the **zero-mode** — corresponds to the "overall translation" degree of freedom shared by all experts. It carries no physical content (all pairwise displacements are invariant under this transformation), but it makes the least-squares problem ill-posed and must be fixed numerically.

4.3 Zero-Mode Fixing: $\sum g_v = 0$ (Not Coulomb Gauge!)

We now arrive at a critical point of clarification.

Theorem 4.1 ($\sum_v g_v = 0$ Is Zero-Mode Fixing, Not Coulomb Gauge)**.** The constraint

$$\sum_{v \in \mathcal{V}} g_v = 0$$

(or equivalently $\langle g, \mathbf{1} \rangle = 0$, i.e., $g \perp \mathbf{1}$) serves to **eliminate the zero-mode to ensure solution uniqueness**. It is **not** the discrete analog of the continuous Coulomb gauge $\partial_\mu A^\mu = 0$.

Reasons:

(1) **Different object being constrained.** $\sum g_v = 0$ is a linear constraint on the **gauge parameter** $g \in \Omega^0$. Coulomb gauge $\partial_\mu A^\mu = 0$ is a divergence constraint on the **gauge potential** $A \in \Omega^1$. The former lives in $\Omega^0(\mathcal{G})$, the latter in $\Omega^1(\mathcal{G})$.

(2) **Different mathematical type.** $\sum g = 0$ is algebraic (g orthogonal to the zero-mode direction). The divergence $d_0^T A = 0$ is differential (requiring discrete divergence to vanish). They act on different spaces.

(3) **Different continuous analog.** The continuous analog of $\sum g_v = 0$ is not $\partial_\mu A^\mu = 0$, but the orthogonality condition $\int \Lambda(x)\, dx = 0$ on the gauge parameter — fixing the integration constant in the gauge transformation $A \to A - d\Lambda$. This fixes the "zero-mode of the gauge parameter," not imposes a condition on the connection.

(4) **Different role in computation.** In solving the normal equation $B^T B g = B^T A$, the constraint $\sum g = 0$ is equivalent to taking the pseudo-inverse or minimum-norm solution on $\mathbf{1}^\perp$ (the orthogonal complement of the all-ones vector). It makes the linear system non-singular without changing the value of the residual $\mathcal{R}[g]$.

Remark 4.2 (What Is the Genuine Discrete Coulomb Gauge?)**.** A genuine discrete Coulomb gauge analog would be:

$$d_0^T A = 0 \quad \text{or} \quad d_0^T (A - d_0 g) = 0,$$

i.e., requiring the discrete divergence of the gauge-fixed edge assignment to vanish. This is an entirely different equation, involving Δ_1 rather than Δ_0, and is **never used** in the SCX computation. The SCX solution g satisfies $B^T B g = B^T A$ together with $\sum g = 0$, but does not guarantee $B^T(A - Bg) = 0$ (in fact the latter *is* the normal equation itself, not an additional condition).

4.4 Solving the Gauge-Fixing Problem

Proposition 4.2 (Closed-Form Solution with Zero-Mode Fixing)**.** With zero-mode fixing constraint $\sum_v g_{v,a} = 0$ (imposed independently for each coordinate dimension), the least-squares problem:

$$\min_{g : \sum_v g_{v,a} = 0} \sum_{a=1}^{d} \|\mathbf{A}_a - B \mathbf{g}_a\|^2$$

has solution given by the pseudo-inverse:

$$\mathbf{g}_a^* = (B^T B)^+ B^T \mathbf{A}_a,$$

where $(B^T B)^+$ is the Moore-Penrose pseudo-inverse of $B^T B$. Equivalently, $\mathbf{g}_a^*$ is the minimum-norm solution, satisfying $\mathbf{g}_a^* \perp \mathbf{1}$ (i.e., automatically satisfying $\sum_v g_{v,a}^* = 0$).

Numerically, one can solve the augmented system:

$$\begin{pmatrix} B^T B & \mathbf{1} \\ \mathbf{1}^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{g}_a \\ \lambda \end{pmatrix} = \begin{pmatrix} B^T \mathbf{A}_a \\ 0 \end{pmatrix}$$

where λ is a Lagrange multiplier.

4.5 Geometric Interpretation of the Solution

Proposition 4.3 (Gauge-Fixing as Orthogonal Projection)**.** The optimal vertex potential g^* satisfies:

$$d_0 g^* = \mathrm{proj}_{\mathrm{im}(B)}(\mathbf{A}),$$

i.e., Bg^* (the vector form of $d_0 g^*$) is the orthogonal projection of $\mathbf{A}$ onto $\mathrm{im}(B)$, the column space of B.

Thus gauge-fixing decomposes the edge assignment A into two parts:

- **Gauge-removable part:** $d_0 g^* \in \mathrm{im}(d_0) \subseteq \Omega^1$ — the component that can be fully absorbed by choosing gauge parameters (the gradient part);
- **Ineliminable residual:** $A - d_0 g^* \perp \mathrm{im}(d_0)$ — the component that cannot be removed by any choice of gauge parameters.

Remark 4.3 (Relation to Hodge Decomposition)**.** The residual $A - d_0 g^*$ lies in $\mathrm{im}(d_0)^\perp$. By the discrete Hodge decomposition (Theorem 2.2): $\mathrm{im}(d_0)^\perp = \ker(\Delta_1) \oplus \mathrm{im}(d_1^T)$. The post-gauge-fixing residual thus contains a **harmonic part** and a **coexact part**, jointly measuring the content of A that is "not removable by a gradient."

In particular, $\dim \ker(\Delta_1)$ equals the cyclomatic number $|\mathcal{L}|$ of the graph — the number of independent elementary loops. The harmonic component precisely corresponds to non-zero loop holonomies — i.e., non-zero discrete curvature.

5 The Cercis Score: Residual Norm after Gauge-Fixing

5.1 Definition: Residual, not Yang-Mills Functional

We now give the **unique definition** of the Cercis Score, one that directly matches the actual SCX computation.

Definition 5.1 (Cercis Score)**.** Let g^* be the optimal vertex potential from the zero-mode-fixed least-squares problem (Proposition 4.2). The **Cercis Score** is defined as the **residual norm after gauge-fixing**:

$$\mathcal{C} := \mathcal{R}[g^*] = \sum_{e \in \mathcal{E}} \|A_e - (d_0 g^*)_e\|^2.$$

Expanding component-wise (per coordinate dimension):

$$\mathcal{C} = \sum_{a=1}^{d} \|\mathbf{A}_a - B\mathbf{g}_a^*\|^2 = \sum_{a=1}^{d} \|\mathbf{A}_a - B(B^T B)^+ B^T \mathbf{A}_a\|^2.$$

Let $P = B(B^T B)^+ B^T$ be the orthogonal projection matrix onto $\mathrm{im}(B)$, and $P^\perp = I - P$ the projector onto $\mathrm{im}(B)^\perp$. Then concisely:

$$\mathcal{C} = \sum_{a=1}^{d} \|P^\perp \mathbf{A}_a\|^2 = \sum_{a=1}^{d} \mathbf{A}_a^T P^\perp \mathbf{A}_a.$$

Remark 5.1 (Cercis Is Not the Yang-Mills Functional)**.** The Cercis Score in this paper is **not** the Yang-Mills functional $\int \|F\|^2$. It is not the sum of squared curvature norms, not the L^2 norm of $d_1 A$. Cercis is the **residual** after gauge-fixing — the squared distance from A to $\mathrm{im}(d_0)$.

The curvature $\boldsymbol{\kappa} = d_1 A$ satisfies $\boldsymbol{\kappa} = d_1(A - d_0 g^*)$ (since $d_1 d_0 = 0$), so curvature is determined solely by the residual $A - d_0 g^*$. But Cercis is the *total* norm of the residual, not just its harmonic component (i.e., curvature). Specifically, if $r = A - d_0 g^*$, then:

$$\mathcal{C} = \|r\|^2 = \|r_{\text{harm}}\|^2 + \|r_{\text{coexact}}\|^2,$$

whereas a Yang-Mills-type functional would only capture $\|d_1 r\|^2 = \|d_1 r_{\text{harm}}\|^2$. These are different, and Cercis is the quantity actually used by SCX.

5.2 Gauge Invariance

Theorem 5.1 (Gauge Invariance of Cercis)**.** The Cercis Score $\mathcal{C}$ is gauge-invariant: under the gauge transformation $A \mapsto A' = A - d_0 h$ (where h is an arbitrary vertex potential), $\mathcal{C}' = \mathcal{C}$.

Proof. Let g^* be the optimal vertex potential for the original A. Under the transformed edge assignment $A' = A - d_0 h$, the optimum shifts to $g'^* = g^* - h$, (since $A' - d_0 g'^* = (A - d_0 h) - d_0(g^* - h) = A - d_0 g^*$). Hence the residual is unchanged: $\|A' - d_0 g'^*\|^2 = \|A - d_0 g^*\|^2$. Therefore $\mathcal{C}' = \mathcal{C}$. □

Remark 5.2 (Intuition for Gauge Invariance)**.** Since $\mathcal{C}$ is defined as the residual norm — the (squared) distance from A to the subspace $\mathrm{im}(d_0)$ — and $\mathrm{im}(d_0)$ is the tangent space of the gauge orbit (in fact the orbit itself, since the group is abelian), the distance is invariant under gauge transformations.

5.3 Discriminative Properties of Cercis

Theorem 5.2 (Characterization of $\mathcal{C} = 0$)**.** The following are equivalent:

(i) $\mathcal{C} = 0$;

(ii) The edge assignment A is exact: there exists g such that $A = d_0 g$;

(iii) $A \in \mathrm{im}(d_0)$, i.e., A consists entirely of the gradient part;

(iv) Perfect global alignment exists: all experts output consistent coordinates at all configurations;

(v) $\mathrm{curv}(\gamma) = 0$ for all quadrilateral elementary loops γ, *and* the **harmonic** component of A is zero.

Note: the second condition in (v) (harmonic component zero) is necessary: $d_1 A = 0$ alone only excludes the harmonic component but does not guarantee that A is exact (see the remark after Theorem 3.1). In the actual SCX computation, if the graph has a non-trivial harmonic component, $\mathcal{C} > 0$ is possible even when curvature is zero.

5.4 Cercis as a Consistency Measure

Remark 5.3 (Code-Paper Correspondence)**.** The Cercis Score defined here as $\mathcal{C} = \|P^\perp A\|^2$ (the residual norm after discrete Hodge gauge-fixing) represents the **theoretical ideal**: a geometric measure of multi-expert inconsistency that is gauge-invariant and mathematically rigorous. The production SCX codebase (`src/scx/`) computes a **practical approximation**: (s) = Q(s) + $\eta(t) \cdot N(s)$, where $isbinaryvoteconsensusamongexpertsand$ is feature-space novelty. These two definitions agree in limiting cases (perfect experts, zero noise) but differ in finite-sample practice. The gauge-fixing pipeline described in §**??** (solving ${}^T Bg = B^T A$) provides the rigorous foundation; the code's $+\eta N$ is a computationally efficient proxy. Closing this gap is an open engineering problem.

Corollary 5.1. The Cercis Score provides an objective measure of gauge consistency of the multi-expert system:

- $\mathcal{C} = 0$: the system is **gauge-consistent**. A set of gauge translations exists such that aligned coordinates are independent of the specific choice of expert or configuration.
- $\mathcal{C} > 0$: the system is **gauge-inconsistent**. The magnitude of the residual measures the "intrinsic friction" that cannot be eliminated by gauge-fixing. The larger $\mathcal{C}$, the more inconsistent the multi-expert system.

A normalized version of Cercis can be defined as:

$$\bar{\mathcal{C}} := \frac{\mathcal{C}}{\|A\|^2},$$

taking values in $[0, 1]$, where 0 corresponds to perfect consistency and 1 corresponds to complete inconsistency (i.e., A is completely orthogonal to $\mathrm{im}(d_0)$).

6 Honest Acknowledgment of Topological Triviality

6.1 All Chern Classes Vanish

We now honestly confront a fundamental fact that prior work has evaded.

Proposition 6.1 (The SCX Bundle Is Topologically Trivial)**.** From the perspective of continuous principal bundles (if one insists on constructing them):

(1) **The structure group $G \cong \mathbb{R}^{Md}$ is contractible.** As a vector space, $\mathbb{R}^{Md}$ can be continuously contracted to the origin. Therefore all its homotopy groups vanish: $\pi_k(G) = 0$ for $k \geq 0$.

(2) **The base space $\mathcal{X} \subset \mathbb{R}^K$ is contractible.** The parameter space is a subset of $\mathbb{R}^K$ (typically a convex set or cube), also contractible.

(3) **The classifying space is trivial.** For a contractible group G, its classifying space BG is weakly contractible. Hence the set of homotopy classes $[\mathcal{X}, BG] = \{*\}$ — there is only one homotopy class. Therefore all G-principal bundles over $\mathcal{X}$ are trivial.

(4) **All characteristic classes are zero.** Chern classes, Pontryagin classes, Euler classes, and all other real characteristic classes are identically zero. Since $\mathcal{X}$ is contractible $\Rightarrow H^k(\mathcal{X}; \mathbb{R}) = 0$ for $k > 0$. The second de Rham cohomology $H^2_{\mathrm{dR}}(\mathcal{X}; \mathbb{R}) = 0$ trivially.

Therefore, there are no topological obstructions. No topological invariants. No non-trivial bundles. The SCX principal bundle (if we were to construct it) is simply the trivial product $\mathcal{P} \cong G \times \mathcal{X}$.

6.2 The Content Is Geometric, Not Topological

Does this weaken the theory? No. Here is why.

Remark 6.1 (The True Distinction: Flat vs. Non-Flat Connections)**.** Topology deals with the question of when a bundle cannot be reduced to the trivial bundle. For SCX, the bundle is **always** trivial — this is uncontroversial.

The real content is **geometric**: given an edge assignment A, does A lie in $\mathrm{im}(d_0)$ (is A flat / exact)? This is a *linear algebra* question about the relationship between A and the subspace $\mathrm{im}(d_0)$, not a topological question.

Even though the bundle is trivial, the edge assignment A can still have non-zero curvature components (harmonic 1-forms). This is precisely the case when $A \notin \mathrm{im}(d_0)$ — the moment gauge inconsistency appears. The phenomenon is geometric (measuring how far A is from $\mathrm{im}(d_0)$), not topological (involving twisted products or transition functions).

Analogy: on $\mathbb{R}^3$ (contractible, trivial), a vector field $\mathbf{v}$ may be curl-free ($\nabla \times \mathbf{v} = 0$, $\mathbf{v} = \nabla\phi$) or not. This is not a topological statement — the topology of $\mathbb{R}^3$ never changes. Similarly, SCX's gauge consistency problem concerns the decomposition of A on the given (trivial) graph, not the non-triviality of a bundle.

6.3 Topological Content in the Graph Perspective

In the graph perspective, the only "topological" content is the Betti numbers of the graph itself.

Proposition 6.2 (Graph Homology)**.** The (real coefficient) homology of $\mathcal{G}_{\mathrm{SCX}}$ is:

- $H_0(\mathcal{G}) \cong \mathbb{R}^c$, where c is the number of connected components (typically 1);
- $H_1(\mathcal{G}) \cong \mathbb{R}^{|\mathcal{L}|}$, where $|\mathcal{L}|$ is the number of elementary loops (the cyclomatic number);
- $H_k(\mathcal{G}) = 0$ for all $k \geq 2$.

$H_1(\mathcal{G}) \cong \ker(\Delta_1)$ is the space of harmonic 1-forms. A non-zero dimensional H_1 means the graph has $|\mathcal{L}|$ independent loops, each capable of carrying non-trivial holonomy. But this is not bundle topology — it is merely the cycle structure of the graph.

7 Numerical Algorithm

7.1 Data Structures and Algorithm Overview

The complete numerical pipeline of SCX gauge-fixing consists of the following steps. Each step directly corresponds to the mathematical structures described above.

[H] SCX Gauge-Fixing and Cercis Computation

Step 1: **Input:** Raw predictions of M experts at N configurations $\{\tilde{x}_m^k \in \mathbb{R}^d \mid k = 1, \dots, N,\ m = 1, \dots, M\}$.

Step 2: **Construct graph $\mathcal{G}_{\mathbf{SCX}}$:**

- Vertices: $n = N \cdot M$, indexed by (k, m);
- Parameter edges: $(k, m) \to (k+1, m)$, totaling $M \cdot (N - 1)$;
- Expert edges: $(k, i) \to (k, j)$ $(i \neq j)$, totaling $N \cdot M \cdot (M - 1)$;
- Total edges: $m = M(N - 1) + NM(M - 1)$.

Step 3: **Build incidence matrix $B \in \mathbb{R}^{m \times n}$:** For each edge e, $B_{e,v_{\text{target}}} = +1$, $B_{e,v_{\text{source}}} = -1$.

Step 4: **Assemble edge assignment vectors $\mathbf{A}_a \in \mathbb{R}^m$:** (for each coordinate $a = 1, \dots, d$)

- Parameter edges: $A_e = \tilde{x}_{m,a}^{k+1} - \tilde{x}_{m,a}^k$;
- Expert edges: $A_e = \tilde{x}_{i,a}^k - \tilde{x}_{j,a}^k$.

Step 5: **Compute graph Laplacian and right-hand side:**

$$L = B^T B \in \mathbb{R}^{n \times n}, \quad \mathbf{b}_a = B^T \mathbf{A}_a \in \mathbb{R}^n.$$

Step 6: **Solve normal equations with zero-mode fixing:** For each $a = 1, \dots, d$, solve:

$$\begin{pmatrix} L & \mathbf{1} \\ \mathbf{1}^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{g}_a \\ \lambda_a \end{pmatrix} = \begin{pmatrix} \mathbf{b}_a \\ 0 \end{pmatrix}.$$

(Or equivalently, use conjugate gradient on the subspace $\mathbf{1}^\perp$ to solve $L\mathbf{g}_a = \mathbf{b}_a$.)

Step 7: **Compute residual and Cercis Score:**

$$\mathcal{C} = \sum_{a=1}^{d} \|\mathbf{A}_a - B\mathbf{g}_a\|^2.$$

Step 8: **Output aligned predictions (optional):**

$$\hat{x}_m^k = \tilde{x}_m^k - g_m^k, \quad \forall k, m.$$

7.2 Relation to the Original MILP

The original SCX implementation used MILP (Mixed-Integer Linear Programming) for gauge-fixing. The least-squares formulation here is equivalent to the *continuous relaxation* of MILP: when all integer constraints are removed, MILP reduces to solving $B^T Bg = B^T A$ with constraint $\sum g = 0$.

If integer constraints are required (e.g., g_m^k restricted to integer lattice points), rounding or branch-and-bound can be applied on top of the least-squares solution. However, for most application scenarios, the continuous least-squares solution already provides gauge-fixing with sufficient precision.

7.3 Computational Complexity

Proposition 7.1 (Complexity Analysis)**.**

- Graph construction: $O(NM^2)$ (each expert edge requires computing a pairwise displacement);
- Laplacian assembly: $O(m) = O(NM^2)$ (non-zero entry insertion);
- Normal equation solve: $O(n^{3/2})$ to $O(n^2)$ depending on solver ($n = NM$, exploiting the sparsity of L, conjugate gradient gives $O(n \cdot \mathrm{nnz})$ per step);
- Cercis computation: $O(md) = O(NM^2 d)$.

For typical parameter settings ($M \sim 10^1$–10^2, $N \sim 10^2$–10^3, $d \leq 3$), the entire pipeline completes in seconds to minutes on modern hardware.

8 Relationship to Prior Work

This section clarifies the relationship between the formulation in this paper and the prior continuous fiber bundle formulation, and honestly identifies the errors in the latter.

8.1 Correspondences

Concept	**Continuous Fiber Bundle (Prior)**	**Graph Hodge Theory (T**
Base space	Base manifold $\mathcal{X}$	Graph $\mathcal{G}_{\mathrm{SCX}}$
Fiber	$G \cong \mathbb{R}^{Md}$	None (graph has no fiber st
Connection	Ehresmann ω (never constructed)	Edge assignment $A \in \Omega^1$ (ra
Gauge transformation	$A \to A - d\Lambda$	$A \to A - d_0 g$
Curvature	$\Omega = d\omega$ (continuous 2-form)	$\boldsymbol{\kappa} = d_1 A$ (loop holonomy)
Coulomb gauge	$\partial_\mu A^\mu = 0$ (corresponds to no SCX computation)	Does not exist in SCX
$\sum g_m = 0$	Misidentified as Coulomb	Correctly identified as zero-
Cercis	Inconsistent: $\int \|F\|^2$ vs. $Q + \eta N$	Unique definition: residual
Topological content	Claimed non-trivial (incorrect)	Honestly admits triviality

8.2 Errors in Prior Formulation

F1: **F1: $\sum g_m = 0$ is not Coulomb gauge.** As discussed above, it is a zero-mode fixing constraint on the gauge parameters, not a divergence constraint on the connection. Calling it Coulomb gauge conflates two entirely different mathematical structures. The correct continuous analog is fixing the integration constant $\int \Lambda = 0$ in the gauge transformation $A \to A - d\Lambda$.

F2: **F2: The connection ω was never constructed.** The continuous framework asserts the existence of a principal bundle with a connection, but never explains how to construct that connection from the pairwise expert displacements δ_{ij}^k. This paper takes the more honest path: δ_{ij}^k *are* the edge assignments A_e — they are the data themselves, requiring no separate "connection" to mediate.

F3: **F3: Cercis definition is inconsistent.** The prior text oscillates between two mutually incompatible definitions: (a) $\mathcal{C} = \int \|F\|^2$ (Yang-Mills functional), (b) $\mathcal{C} = Q + \eta N$ (the $Q + \eta N$ actually used by SCX). This paper unifies them as the residual norm $\mathcal{C} = \|P^\perp A\|^2$, which agrees with (b) and is not confused with (a).

F4: **F4: Topology is misrepresented.** The prior text suggests non-trivial topological structure, but G contractible and $\mathcal{X}$ contractible $\Rightarrow$ all Chern classes vanish $\Rightarrow$ the bundle is trivial. This paper acknowledges this explicitly and correctly interprets the content as geometric (flat vs. non-flat), not topological.

9 Discrete Hodge vs. Continuous Differential Geometry

This section provides a systematic comparison between the discrete graph Hodge method and the continuous differential geometry (fiber bundle) method, delineates their respective capabilities, and explains why the discrete method is the correct choice for SCX.

9.1 Two Formalisms

Capability	**Discrete Hodge (This Paper)**	**Continuous Fiber Bundle (Prior)**
Direct construction from raw data	✓ Edge assignment A_e is the data	× Connection ω never constructed
Precise definition of curvature	✓ Loop holonomy $d_1 A$	✓ Curvature 2-form Ω
Gauge transformation formula	✓ $A' = A - d_0 g$	✓ $A' = A - d\Lambda$
Gauge-fixing algorithm	✓ Least squares $B^T B g = B^T A$	× No corresponding discrete algorithm
Unique definition of Cercis	✓ $\|P^{\perp} A\|^2$	× Oscillates between $\int \|F\|^2$ and $Q + \eta N$
Zero-mode handling	✓ $\sum g_v = 0$ correctly identified	× Misidentified as Coulomb gauge
Topological content	✓ Honestly admits triviality	× Suggests non-trivial topology
Numerical computability	✓ Sparse linear system solve	× Continuous PDE, not discretizable
Consistency with SCX code	✓ Line-by-line correspondence	× Loose analogy only
Path to non-abelian generalization	✓ Group-valued edge assignments + optimization	✓ Principal bundle + connection
Connection to persistent homology	✓ H_1 barcode	× No natural connection
Infinite-dimensional limit	✓ Graph limits to mesh	✓ Discretization → continuous PDE

9.2 What Continuous Differential Geometry Can Do

Continuous differential geometry (fiber bundles, connections, curvature) is powerful in the following respects:

(1) **Global topological classification.** When the base manifold and structure group are non-trivial, Chern–Weil theory provides a complete classification of topological obstructions via characteristic classes. For example, the instanton number of an

$SU(2)$-bundle over S^4 is given by the integral of the second Chern class c_2, and must be an integer. This entire machinery is irreplaceable for physical systems with non-trivial topology.

(2) **Analytical tools.** The Atiyah–Singer index theorem relates the analytic index of elliptic operators to topological invariants. Hodge theory identifies the space of harmonic forms with de Rham cohomology. These deep theorems provide bridges between analytic computation and topological classification.

(3) **Local-to-global integration.** The connection 1-form unifies local gauge transformations ($A \to A - d\Lambda$) with global parallel transport. Wilson lines $W[\gamma] = \mathcal{P}\exp(\oint_\gamma A)$ are gauge-invariant and encode path-dependent information. This structure is indispensable in the non-abelian case.

(4) **Mature theoretical ecosystem.** Six decades of development have produced a complete body of theorems (Chern–Weil, Atiyah–Singer, Witten anomaly, Seiberg–Witten), from which any gauge theory problem can draw insight.

9.3 What Continuous Differential Geometry Cannot Do

However, for the specific problem of SCX, continuous differential geometry has the following fundamental limitations:

(1) **Cannot construct a connection from discrete data.** The continuous framework requires a smooth connection 1-form ω on a base manifold. But SCX's raw data consist of **finitely many** pairwise displacements δ_{ij}^k — a set of discrete numbers. Constructing a smooth connection from finitely many point values is an **ill-posed inverse problem**: infinitely many smooth connections can interpolate the given discrete values, with no uniqueness principle to select one. Prior work never explicitly constructed a connection — it merely *asserted* existence.

(2) **All topological invariants are identically zero.** SCX's base space $\mathcal{X} \subset \mathbb{R}^K$ is contractible (typically a convex parameter region), and the structure group $G \cong \mathbb{R}^{Md}$ is also contractible. Therefore:

- $H^k(\mathcal{X}; \mathbb{R}) = 0$ for all $k > 0$;
- All Chern classes $c_k(P) \in H^{2k}(\mathcal{X}; \mathbb{R}) = 0$ vanish identically;
- The principal bundle is globally trivial: $P \cong G \times \mathcal{X}$;
- The set of homotopy classes $[\mathcal{X}, BG] = \{*\}$ has a single element.

This means the entire heavy weaponry of Chern–Weil theory, index theorems, and instanton classification **produces no non-trivial output here**. All topological invariants they provide are zero. Citing these theorems generates no new information about the SCX system.

(3) **The continuous formalism is disconnected from the computation.** The actual computation of the Cercis Score is: solve the sparse linear system $B^T B g = B^T A$ for $M \times N$ vertices (or equivalently, compute the projection of A onto $\mathrm{im}(B)$). The continuous framework describes this as minimization of the Yang–Mills functional $\int \|F\|^2$, but these two quantities are **not the same mathematical object**:

- $\int \|F\|^2$ is the L^2 norm of the curvature 2-form;
- $\|P^\perp A\|^2$ is the squared distance from the edge assignment to the gradient subspace;
- The former captures only the harmonic component (curvature), while the latter contains both harmonic and coexact parts.

Confusing them is a category error.

(4) **Terminological confusion.** The continuous framework calls $\sum g_v = 0$ the "Coulomb gauge," but Coulomb gauge is $\partial_\mu A^\mu = 0$ — a divergence constraint on the gauge **potential**. $\sum g_v = 0$ is zero-mode fixing on the gauge **parameter**. Confusing these two concepts is not merely imprecise — it is incorrect. A physically trained referee would immediately recognize this error.

9.4 Why Discrete Is the Right Choice

The advantages of the discrete graph Hodge method stem from the following principles:

(1) **Ontological Honesty.** SCX's data are discrete. The graph is discrete. The computation is discrete. Discrete Hodge theory **is** the native mathematics of the data — no interpolation, no smoothing, no unverifiable step of lifting discrete numbers to continuous fields is needed.

(2) **Computational Correspondence.** Every mathematical object in this paper directly corresponds to one or more lines in the SCX code:

- $A_e = \delta^k_{ij}$: direct storage of pairwise expert displacements;
- B: the adjacency structure of the graph, used to build the normal equations;
- $B^T B g = B^T A$: the exact form of the continuous relaxation of MILP;
- $\mathcal{C} = \|P^\perp A\|^2$: direct computation of the residual after gauge-fixing.

(3) **Mathematical Self-Sufficiency.** Discrete Hodge theory is a complete, self-contained mathematical system. It has its own Hodge decomposition, its own Laplacians, its own cohomology. It does not need to reference continuous theory for self-justification. In fact, continuous de Rham cohomology can be *recovered* by taking an appropriate limit of discrete Hodge theory, not the other way around.

(4) **Falsifiability.** Every statement in the discrete formulation can be directly verified by numerical computation. Statements in the continuous formulation (such as "there exists a connection ω") are typically unverifiable or unfalsifiable — because they provide no construction.

(5) **Natural Generalization Path.** The discrete framework provides clear paths for generalization:

- Weighted graphs: W_0, W_1 weight matrices;
- Non-abelian group-valued edge assignments: group-valued least squares;
- Persistent homology: Cercis as H_1 persistence;
- Infinite-dimensional limit: graph $\to$ mesh $\to$ PDE.

None of these generalizations require first passing through the continuous framework.

9.5 What the Continuous Formalism Would Need to Become Valid

If future work insists on using the continuous differential geometry framework, at minimum the following steps must be completed:

C1: **C1: Explicitly construct a connection from discrete data.** An algorithm must be given that explicitly constructs a smooth connection 1-form ω from finitely many pairwise displacements $\{\delta_{ij}^k\}$ (e.g., via interpolation or reconstruction). Merely asserting that such a connection exists is insufficient.

C2: **C2: Prove consistency of discretization.** It must be proved that as the discrete data are refined (number of configurations $N \to \infty$ and number of experts $M \to \infty$), the discrete edge assignments A_e converge, in some appropriate sense, to the continuous connection ω. This is analogous to proving the existence of the continuum limit in lattice gauge theory.

C3: **C3: Establish non-trivial topology.** Non-trivial topological structure must be identified that *actually exists* in the base space or structure group. If both the base space and the group are contractible, one must acknowledge that the topological part of the continuous framework produces no non-trivial output. Alternatives include considering non-trivial group actions or parameter spaces with holes.

C4: **C4: Precisely relate Cercis to the Yang-Mills functional.** If it is claimed that the Cercis Score corresponds to $\int \|F\|^2$, one must mathematically prove that they are equal, or at least converge in an appropriate limit. Currently these two quantities are different mathematical objects in different spaces.

C5: **C5: Correctly define the Coulomb gauge analog.** A correct discretization of the continuous Coulomb gauge $\partial_\mu A^\mu = 0$ must be given, and it must be distinguished from the constraint $\sum g_v = 0$ actually used by SCX. If they are different (as they indeed are), one must state under what conditions they are equivalent, or acknowledge that they are unrelated.

Summary. Discrete graph Hodge theory and continuous fiber bundle theory are not competitors — they are formalisms at different levels, suited to different problems. For SCX, the data are naturally discrete, the computation is naturally discrete, and therefore the discrete formalism is the **only formalism consistent with the computation**. The continuous framework can only be considered a valid alternative after completing the above steps C1–C5. Until then, discrete Hodge theory should be regarded as the **authoritative formulation** of SCX gauge theory.

10 Generalizations and Outlook

10.1 Weighted Graphs and More General Metrics

In this paper we used unweighted graphs (all edge weights equal). More generally, edge weights can be assigned to reflect confidence in certain expert or configuration comparisons:

$$\mathcal{R}_W[g] := \sum_{e \in \mathcal{E}} w_e \, \|A_e - (d_0 g)_e\|^2.$$

The weighted Laplacian $L_W = B^T W B$ replaces the unweighted version, and all theoretical results carry through unchanged.

The weighted version allows tuning Cercis to be more sensitive to important expert pairs or critical configuration regions.

10.2 Non-Abelian Expert Symmetries

If the gauge group of experts extends to include rotations and scaling (e.g., $\mathrm{SE}(d)$), the edge assignments become group-valued rather than vector-valued. In this case:

- Gauge transformations g are group-valued vertex functions;
- Edge assignments transform as $A'_e = g_v \circ A_e \circ g_u^{-1}$;
- Gauge-fixing becomes a group-valued least-squares problem, requiring solution on homogeneous spaces (using optimization methods on Riemannian manifolds);
- Curvature can still be defined as the group product along loops.

The graph-theoretic framework of this paper provides a natural foundation for such generalizations.

10.3 Connection to Persistent Homology

The Cercis Score can be interpreted as a persistence measure in persistent homology:

- Non-zero discrete curvature loops correspond to 1-dimensional “holes” in the barcode;
- The harmonic component of Cercis, $\|r_{\text{harm}}\|^2$, can be viewed as a persistence-weighted sum of H_1 classes under some filtration;
- Gauge-fixing is equivalent to searching for the optimal coboundary to “fill” these 1-dimensional holes.

10.4 Infinite-Dimensional Limit

As $M \to \infty$ and $N \to \infty$, the graph $\mathcal{G}_{\text{SCX}}$ tends to a continuous grid on the product space $[0,1] \times [0,1]$. The limiting PDE involves derivatives in both parameter and expert directions, with natural connections to 2D gauge field theory. Rigorous analysis of this limit is ongoing.

11 Conclusion

11.1 Main Results

This paper establishes a complete discrete formalization of SCX gauge theory on graphs. The main contributions are:

(1) **Graph Hodge framework:** The mathematical structure of SCX is precisely located as discrete Hodge theory on directed graphs. Pairwise expert displacements are directly encoded as edge assignments A_e — no continuous connection or fiber bundle needed.

(2) **Curvature as loop holonomy:** Discrete curvature is defined as the algebraic sum of edge assignments around elementary quadrilateral loops. Curvature is nonzero iff experts' predictions change inconsistently under parameter variation — the geometric formulation of PES misalignment.

(3) **Gauge-fixing as least squares:** Gauge-fixing is rigorously formalized as finding vertex potentials g minimizing $\sum_e \|A_e - (g_w - g_v)\|^2$.

(4) **Zero-mode fixing $\neq$ Coulomb gauge:** Strictly distinguishes the constraint $\sum_v g_v = 0$ (zero-mode fixing, ensuring solution uniqueness) from Coulomb gauge $\partial_\mu A^\mu = 0$ (divergence constraint on the connection). The former is algebraic, the latter differential — they are entirely different.

(5) **Unique Cercis definition:** The Cercis Score is defined as the residual norm $\|P^\perp A\|^2$ after gauge-fixing, i.e., the squared distance from the edge assignment to the gradient subspace. This definition is unique, gauge-invariant, and precisely matches the actual SCX computation.

(6) **Honest topological triviality:** Explicitly acknowledges that G is contractible and $\mathcal{X}$ is contractible, hence the bundle is topologically trivial. All content is geometric (flat vs. non-flat edge assignments), not topological.

11.2 Final Remark

The mathematical core of SCX is elegant and simple: it is just discrete Hodge decomposition on a graph. Given a set of pairwise expert displacements (edge assignment A), the problem reduces to determining the distance between A and the gradient subspace $\mathrm{im}(d_0)$. Gauge consistency means this distance is zero (A is exact); inconsistency means it is positive, and Cercis precisely quantifies that distance.

The introduction of continuous differential geometry — fiber bundles, connections, curvature forms — is superfluous for this problem and even harmful, because it introduces terminology (Coulomb gauge, Yang-Mills action, Chern classes) that does not correspond to the actual discrete computation.

We advocate: a clear, honest, and mathematically precise discrete formulation is always preferable to an inaccurate continuous analogy disconnected from the computation.

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